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TREND-BASED MOCK PAPER

GATE Architecture and Planning

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Architecture and Planning
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

*Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.*

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Scale 1:100, drawing 10cm > actual:

(Numerical Answer Type — Enter the numerical value)

Section B: Architecture and Planning

Q2. [1-NAT] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
 - (B) 0.5
 - (C) 0.6
 - (D) 0.4
-

Q4. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
 - (B) 2
 - (C) 4
 - (D) 1
-

Q5. [1-MCQ] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1

- (B) -2
- (C) 1
- (D) 2

Q6. [1-MCQ] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q7. [1-MCQ] [1 mark]

The vector (1,2,3) projected onto (2,1,0) is:

- (A) (4/5, 2/5, 0)
- (B) (8/5, 4/5, 0)
- (C) (2, 1, 0)
- (D) (1, 2, 3)

Q8. [1-NAT] [1 mark]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q10. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q11. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q12. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q13. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q14. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q15. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q16. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q17. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q18. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q19. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q20. [2-MSQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q21. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q22. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q23. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q25. [2-MSQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q26. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q27. [2-NAT] [2 marks]

The Laplace transform of $t e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4

- (B) 8
- (C) 16
- (D) 12

Q29. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q30. [2-MCQ] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q31. [1-NAT] [1 mark]

Slump test measures:

(Numerical Answer Type — Enter the numerical value)

Q32. [1-MCQ] [1 mark]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q33. [1-MCQ] [1 mark]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q34. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q35. [2-NAT] [2 marks]

M25 concrete modulus of elasticity:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-MCQ] [2 marks]

Camber on road provides:

- (A) Speed
- (B) Drainage

- (C) Aesthetics
- (D) Strength

Q37. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q38. [2-NAT] [2 marks]

M25 concrete modulus of elasticity:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q40. [2-MCQ] [2 marks]

M25 concrete modulus of elasticity:

- (A) 25000 MPa
- (B) 5000 MPa
- (C) 50000 MPa
- (D) 10000 MPa

Q41. [1-MCQ] [1 mark]

Kath-Kuni from:

- (A) Rajasthan
- (B) Himachal Pradesh
- (C) Kerala
- (D) Gujarat

Q42. [1-NAT] [1 mark]

Taj Mahal built by:

(Numerical Answer Type — Enter the numerical value)

Q43. [1-MCQ] [1 mark]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q44. [1-MCQ] [1 mark]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6

(D) 1:1:2

Q45. [2-MCQ] [2 marks]

Taj Mahal built by:

- (A) Akbar
- (B) Shah Jahan
- (C) Aurangzeb
- (D) Humayun

Q46. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q47. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q48. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q49. [2-NAT] [2 marks]

Taj Mahal built by:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

M20 concrete mix ratio:

- (A) 1:1.5:3
- (B) 1:2:4
- (C) 1:3:6
- (D) 1:1:2

Q51. [1-MCQ] [1 mark]

Cantilever UDL w, L. Fixed end moment:

- (A) $wL^2/2$
- (B) wL^2
- (C) $wL^2/4$
- (D) $wL^2/8$

Q52. [1-NAT] [1 mark]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-NAT] [1 mark]

Influence line for SS beam reaction:

(Numerical Answer Type — Enter the numerical value)

Q54. [2-MCQ] [2 marks]

Influence line for SS beam reaction:

- (A) Triangle
 - (B) Rectangle
 - (C) Linear decreasing
 - (D) Parabola
-

Q55. [2-MCQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
 - (B) 0.43
 - (C) 0.5
 - (D) 0.75
-

Q56. [2-MCQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
 - (B) 0.43
 - (C) 0.5
 - (D) 0.75
-

Q57. [2-MSQ] [2 marks]

Influence line for SS beam reaction:

- (A) Triangle
 - (B) Rectangle
 - (C) Linear decreasing
 - (D) Parabola
-

Q58. [2-MCQ] [2 marks]

Moment distribution DF for $K_{AB}=4EI/L$, $K_{AC}=3EI/L$:

- (A) 0.57
 - (B) 0.43
 - (C) 0.5
 - (D) 0.75
-

Q59. [1-MCQ] [1 mark]

Trap in plumbing:

- (A) Prevents smell
- (B) Stores water
- (C) Both
- (D) None

Q60. [1-NAT] [1 mark]

Sun path in northern hemisphere:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

M20 concrete ratio:

- (A) 1:1:2
 - (B) 1:1.5:3
 - (C) 1:2:4
 - (D) 1:3:6
-

Q62. [1-MCQ] [1 mark]

Zoning regulates:

- (A) Land use
 - (B) Building height
 - (C) Both
 - (D) None
-

Q63. [1-MCQ] [1 mark]

Golden ratio approximately:

- (A) 1.414
 - (B) 1.618
 - (C) 2.0
 - (D) 3.14
-

Q64. [1-NAT] [1 mark]

Load-bearing wall thickness:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-MCQ] [1 mark]

Hardscape includes:

- (A) Plants
 - (B) Pavement
 - (C) Water
 - (D) Trees
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1MCQ	B	1
3	1NAT	B	1
4	1MCQ	B	1
5	1MCQ	C	1
6	1MCQ	B	1
7	1NAT	A	1
8	1MCQ	C	1
9	1NAT	A	1
10	1MCQ	B	1
11	1MCQ	B	1
12	1MCQ	A	1
13	1MCQ	A	1

14	1NAT	B	1
15	1NAT	A	1
16	1NAT	A	1
17	1MCQ	B	1
18	1MCQ	B	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1NAT	C	1
22	1MCQ	A	1
23	1MCQ	A	1
24	1MCQ	B	1
25	1NAT	A	1
26	1MCQ	A	1
27	1MCQ	B	1
28	1MCQ	C	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2MCQ	A	2
32	2MCQ	B	2
33	2MCQ	C	2
34	2MCQ	A	2
35	2MCQ	C	2
36	2MCQ	A	2
37	2MSQ	B	2
38	2MCQ	A	2
39	2NAT	A	2
40	2NAT	C	2
41	2MCQ	A	2
42	2MSQ	B	2
43	2MCQ	A	2

44	2MCQ	C	2
45	2MCQ	B	2
46	2MCQ	B	2
47	2NAT	A	2
48	2MCQ	A	2
49	2MCQ	A	2
50	2MCQ	A	2
51	2NAT	A	2
52	2MCQ	C	2
53	2MCQ	B	2
54	2MCQ	A	2
55	2MCQ	B	2
56	2MCQ	A	2
57	2MCQ	A	2
58	2MCQ	B	2

59	2NAT	B	2
60	2NAT	A	2
61	2MSQ	C	2
62	2MCQ	B	2
63	2MCQ	A	2
64	2MCQ	A	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: difficult

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: Vernacular | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Architecture | Topic: History | Difficulty: difficult

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Building Materials | Topic: Concrete | Difficulty: easy

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Urban Planning | Topic: Zoning | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Design | Topic: Principles | Difficulty: difficult

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Technology | Topic: Structures | Difficulty: easy

Q8.

Answer: C

Solution:

Correct Answer: C

Subject: Climate & Services | Topic: Plumbing | Difficulty: difficult

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Climate & Services | Topic: Solar | Difficulty: moderate

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Landscape Architecture | Topic: Elements | Difficulty: moderate

Q11.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q12.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q13.

Answer: A

Solution:

FEM = $wL^4/(24EI)$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q14.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q15.

Answer: A

Solution:

DF_{AB} = $4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q16.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q17.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q18.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q19.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q20.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q21.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q22.

Answer: A

Solution:

$$E_c = 5000 \sqrt{f_{ck}} = 5000 \sqrt{5} = 25000 \text{ MPa.}$$

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q23.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: B

Solution:

$$\text{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q25.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \text{ Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q26.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q27.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q28.

Answer: C

Solution:

$$\text{Kirchhoff: for } K_n, \text{ spanning trees} = n^{n-2} = 4^2 = 16.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1, 0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1, 0]^T = [1, 0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: B

Solution:

$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q44.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q45.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: difficult

Q46.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q47.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000$ MPa.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q48.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: difficult

Q49.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q50.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: moderate

Q51.

Answer: A

Solution:

$L\{t^s f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q54.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q55.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: difficult

Q56.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q57.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q58.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: B

Solution:

Taj Mahal: built by Mughal emperor Shah Jahan (1632-1653) in memory of Mumtaz Mahal.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q60.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q61.

Answer: C

Solution:

$R_A = (L-x)/L$. Linear decreasing from L to 0.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q62.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q63.

Answer: A

Solution:

$DF_{AB} = 4/(4+3) = 4/7 = 0.57$.

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: moderate

Q64.

Answer: A

Solution:

M20: 1:1.5:3 (cement:sand:aggregate). Characteristic strength = 20 MPa.

Subject: Architecture | Topic: Architecture | Difficulty: easy

Q65.

Answer: A

Solution:

$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}$.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 29 questions, 46 marks
Civil Engineering: 10 questions, 17 marks
Architecture: 10 questions, 16 marks
Structural Analysis: 8 questions, 13 marks
Climate & Services: 2 questions, 2 marks
General Aptitude: 1 questions, 1 marks
Building Materials: 1 questions, 1 marks
Urban Planning: 1 questions, 1 marks
Design: 1 questions, 1 marks
Construction Technology: 1 questions, 1 marks
Landscape Architecture: 1 questions, 1 marks

Difficulty Distribution:

Easy: 19 | Moderate: 34 | Difficult: 12

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.